

Practical Python Foundations

Controlled RAG Evaluation Corpus - Book 1

Purpose

This synthetic mini-book is designed for a beginner-friendly RAG engineering laboratory. It contains stable facts, overlapping terminology, and controlled distractors so retrieval and generation changes can be evaluated reproducibly.

Corpus rule: keep this file unchanged once the baseline evaluation dataset has been created.

Chapter 1: 1. Python's Execution Model

Python programs are organized into modules. A module is normally a .py file whose top-level statements execute when the module is first imported. Python caches imported modules in `sys.modules`, so an ordinary repeated import in the same process normally does not execute the module again.

Names are references to objects. Assignment binds a name to an object rather than copying the object automatically. This distinction explains why two variables can refer to the same mutable list and observe each other's changes.

Functions create local scopes. Parameters are local names bound to objects supplied by the caller. Mutable arguments can be modified by a function, although rebinding a parameter does not rebind the caller's variable.

Chapter 2: 2. Functions, Classes, and Dataclasses

A function should usually perform one clear responsibility. Type hints describe expected types but are not automatically enforced by the Python runtime. Static analysis tools can use those hints to detect likely mistakes.

A class combines data and behavior. An instance method receives the instance through the conventional name `self`. A dataclass can generate common methods such as `__init__` and `__repr__`, making it useful for data-oriented objects.

Default mutable arguments are a common pitfall. A function such as `def add(item, items=[])` reuses the same list across calls. A safer pattern uses `None` and creates a new list inside the function.

Chapter 3: 3. Exceptions and Resource Management

Exceptions separate normal control flow from error handling. Code that can fail belongs in a try block, expected failures can be handled with `except`, `else` runs when no exception occurred, and finally runs whether or not an exception was raised.

The `raise` statement creates or propagates an exception. Custom exception types are useful when callers need to distinguish domain failures from generic programming errors.

A context manager controls setup and cleanup around a block. The `with` statement is commonly used for files because the file is closed even when an exception occurs. Context managers are also useful for database transactions, locks, and temporary resources.

Chapter 4: 4. Testing and Project Tooling

`pytest` discovers tests and provides concise assertions and fixtures. A good unit test checks a small behavior in isolation and should produce the same result when repeated under the same conditions.

A virtual environment isolates project packages from the global Python installation. `pip` installs packages, while a `requirements` file can record package dependencies for reproducible setup.

Environment variables are appropriate for secrets such as API keys. Secrets should not be committed to source control. A .gitignore file commonly excludes .env files, virtual environments, caches, and local databases.

Chapter 5: 5. Search-Relevant Distractors

The word vector in ordinary Python often refers informally to a list or numeric array. This is different from an embedding vector used in semantic search, although both can be represented as sequences of numbers.

Python's import cache is unrelated to application-level response caching. Import caching avoids re-executing modules; response caching stores outputs so repeated requests can avoid expensive work.

Similarity is not a built-in property of Python strings. Semantic similarity normally requires an external representation such as embeddings, whereas lexical comparisons operate on characters or tokens.

Evaluation Notes

This book is synthetic and intentionally compact. Facts are written explicitly so they can support reference answers, retrieval-context checks, groundedness tests, citation experiments, and regression tests. Some passages are deliberate distractors and are labeled as such.